SONATRACH

NATIONAL SCHOOL
OF ARTIFICIAL INTELLIGENCE

***Data mining project*:**

Using Data Mining Techniques to Estimate Well Productivity for Sonatrach's Exploration Division.

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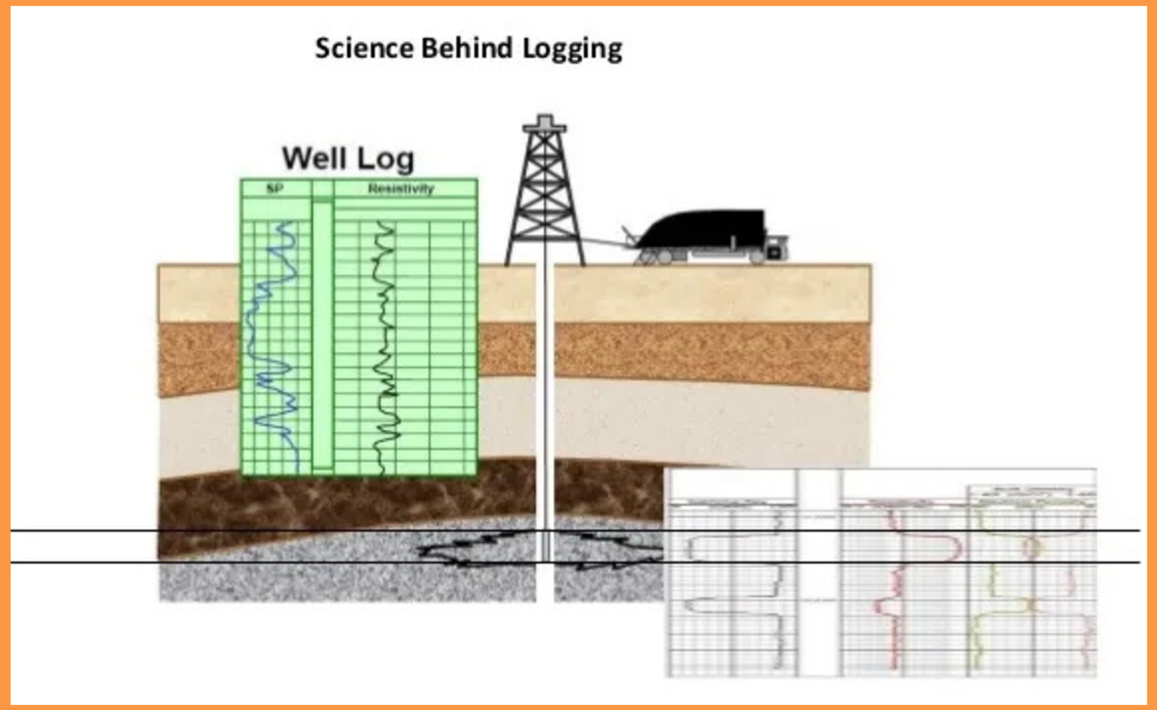
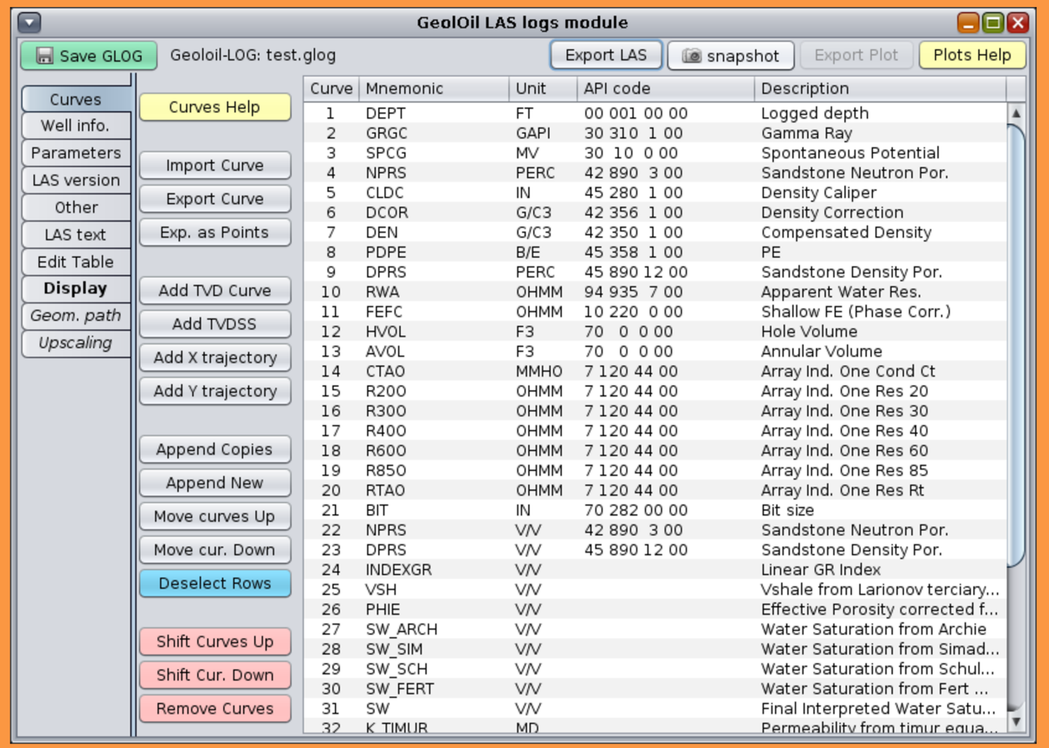
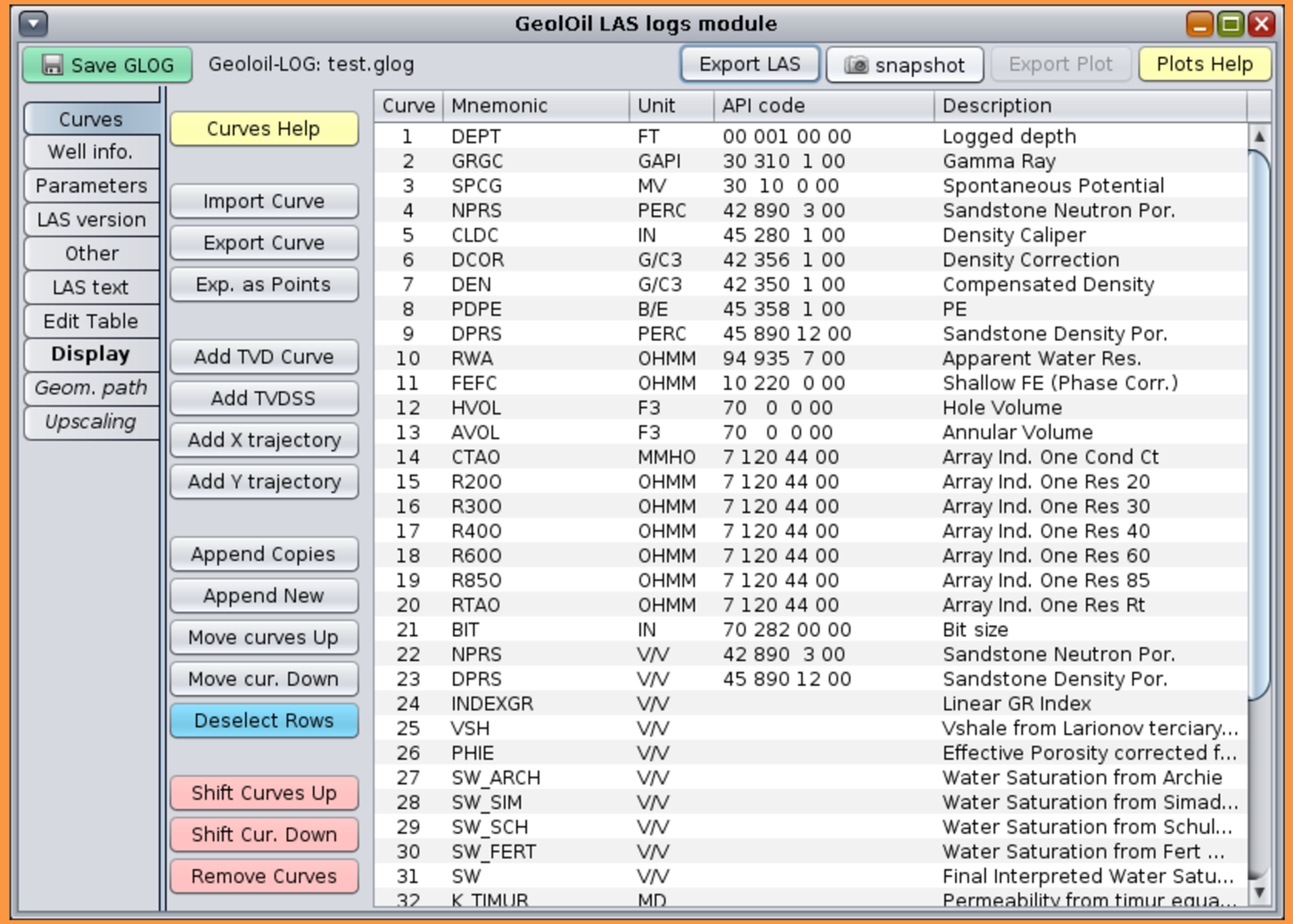
Well Logging => LAS file

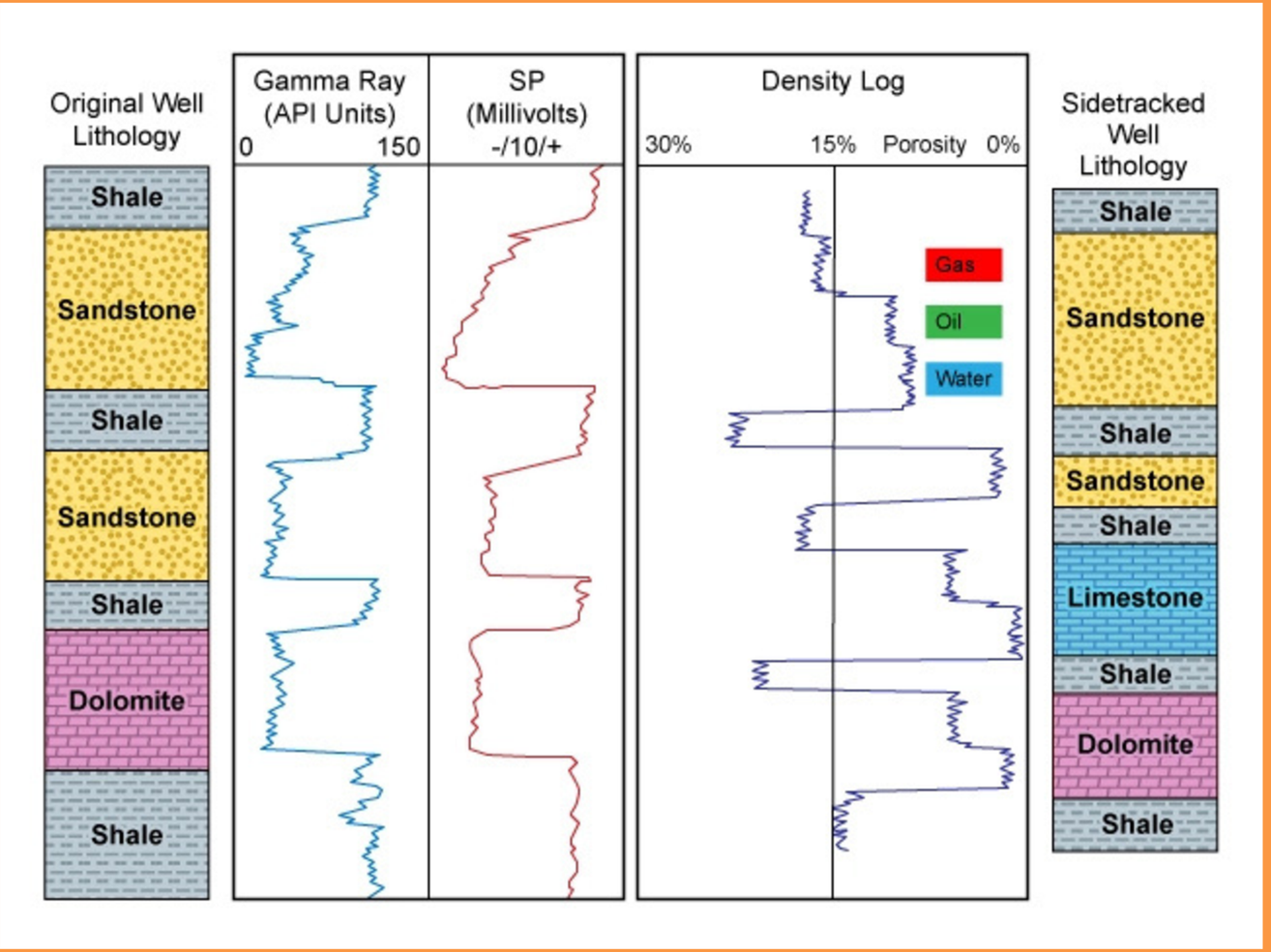






Log Data for each DEPTH

Our Objective ?



PROCESS

1

Estimate the values of the cement that we need to put in the hole to optimize the operation.

2

Time series study, to estimate the values of GR, NPHI, Delta T in the next layers using the given layer.

3

Clustering approach to estimate the lithology at given a depth.

4

Regression approach to estimate the values of Delta T given the others attributes

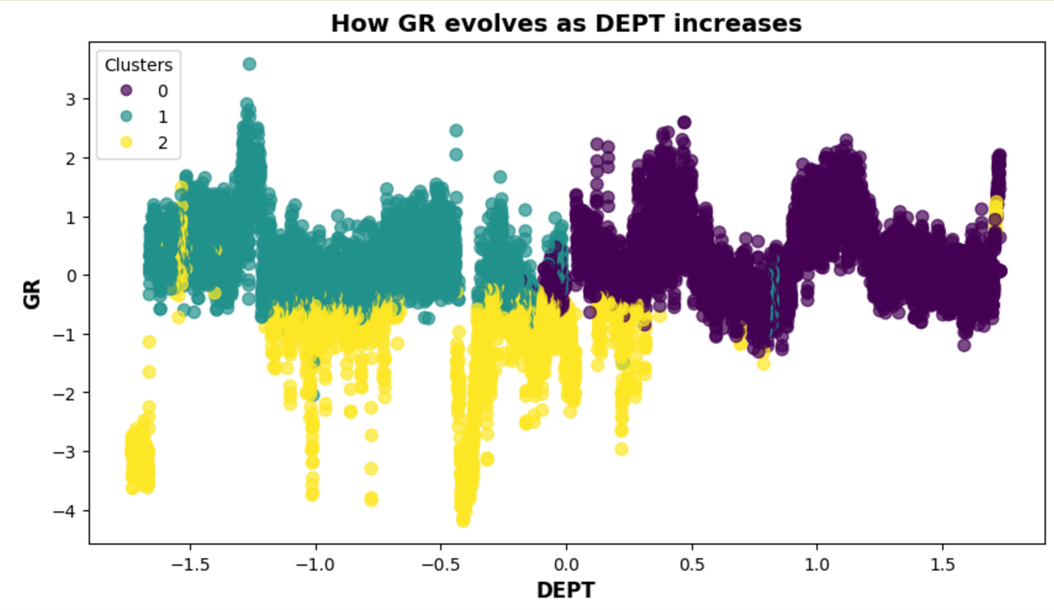
FIRST DATASET :

Data cleaning and preprocessing :

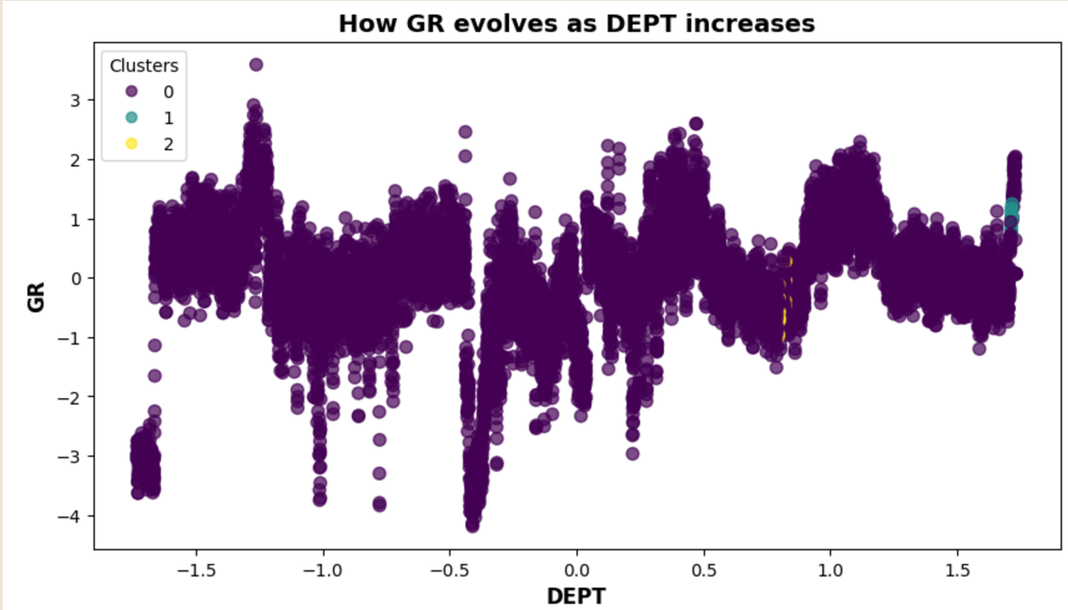
- **High Correlation observed.**
- **Attributes describing the hole characteristics**

Clustering approach :

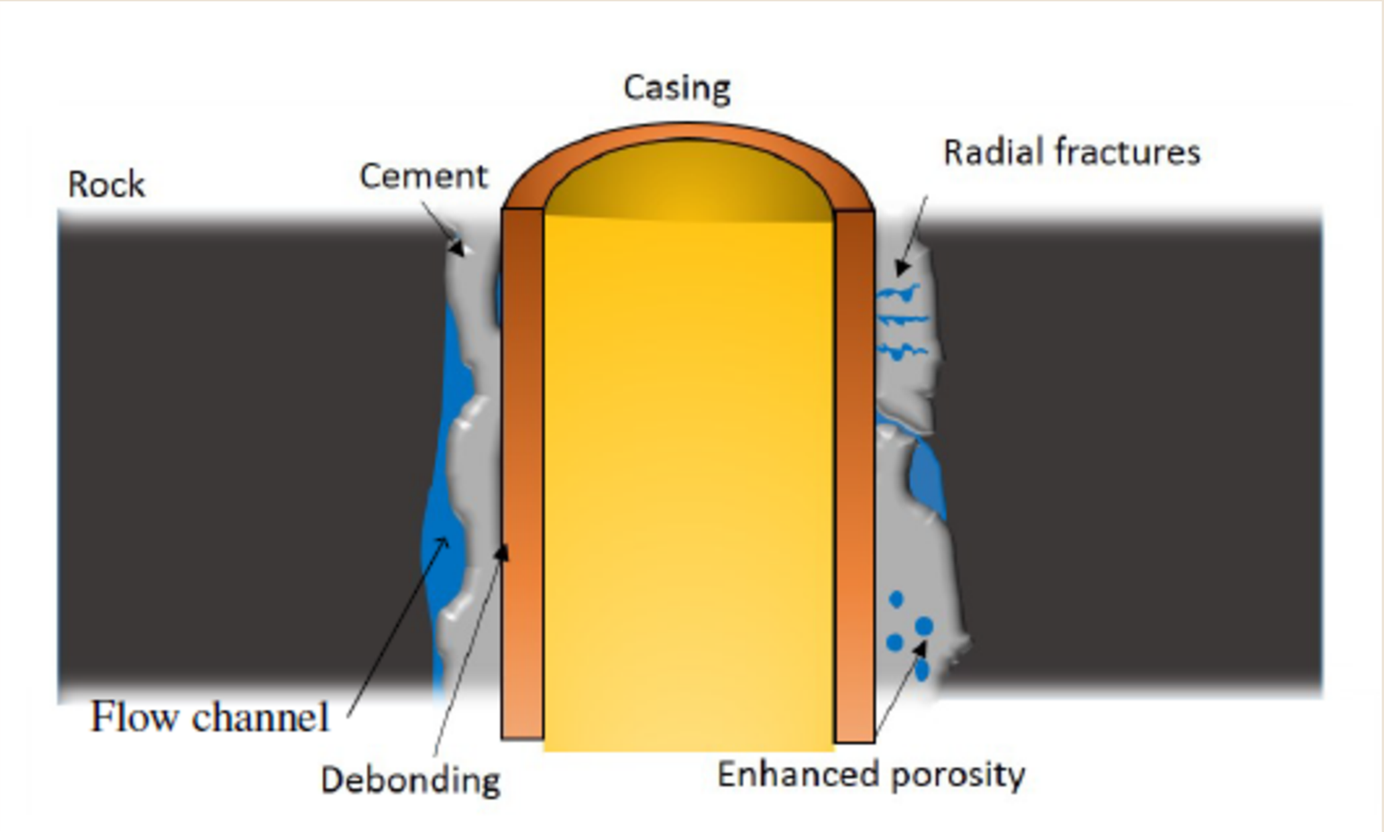
K Means



DBSCAN



ICV estimation approach :



Why care about the cement volume prediction ?

Resources Optimization ?

SECOND DATASET :

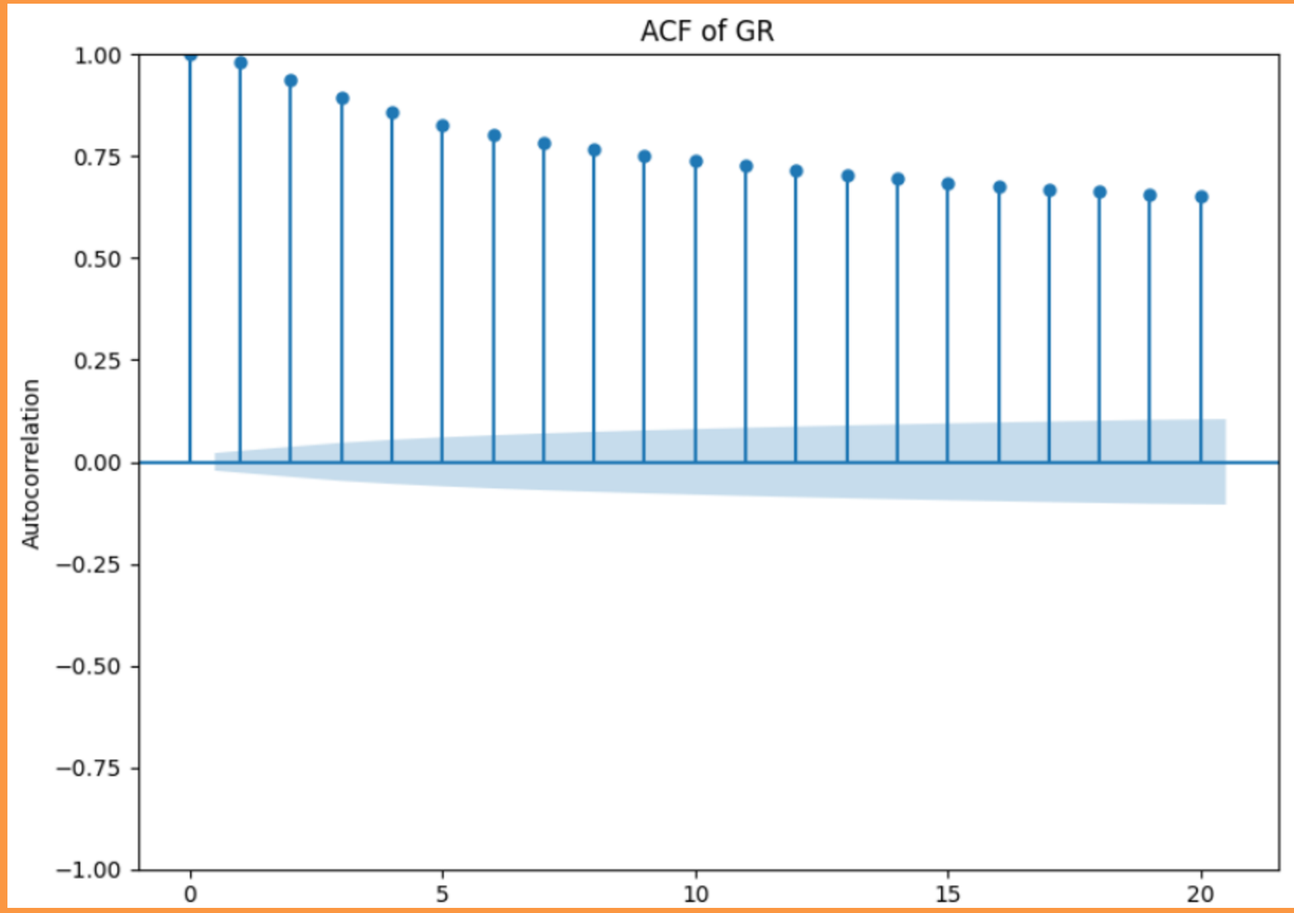
Data cleaning & preprocessing :

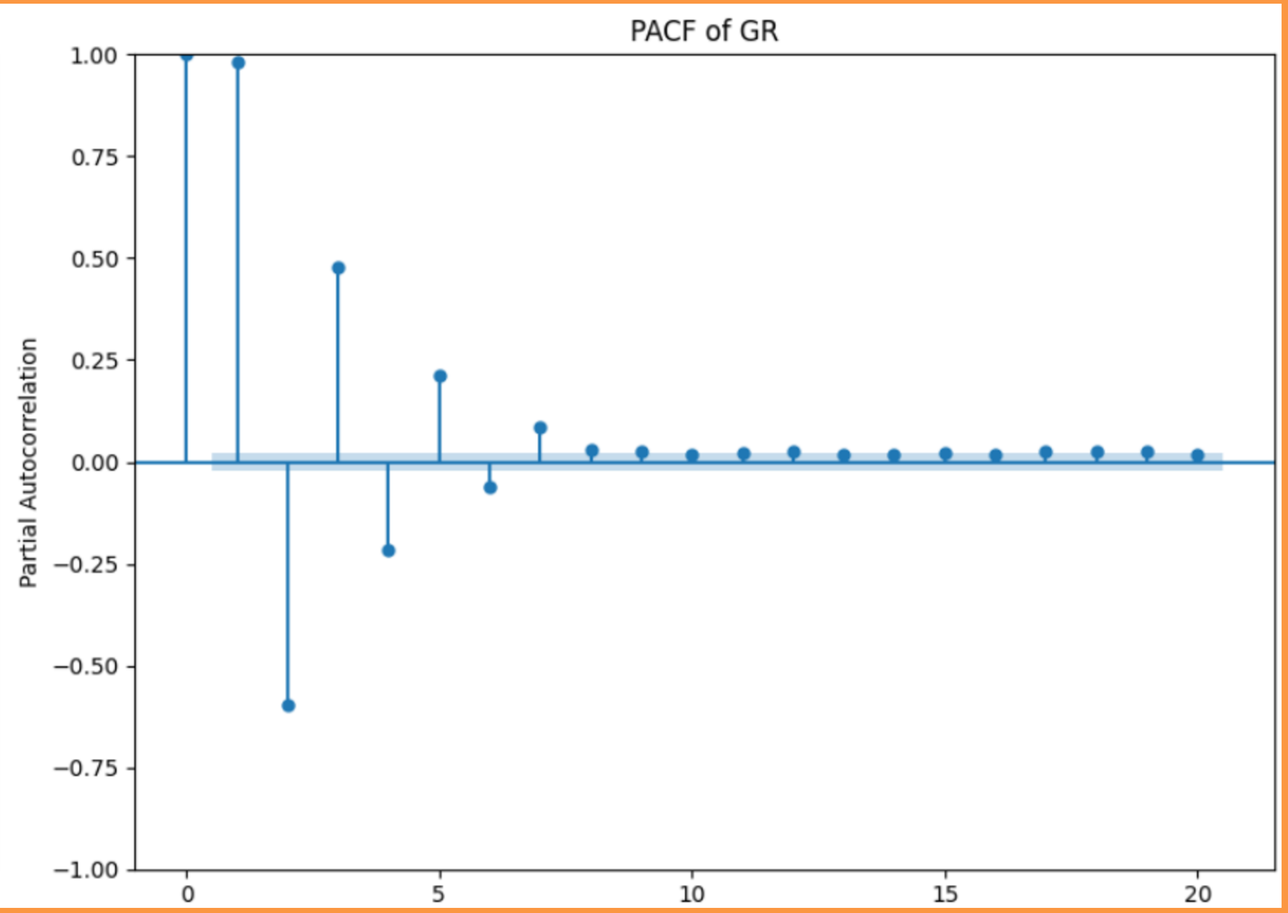
what we did with the negative values in the data ?

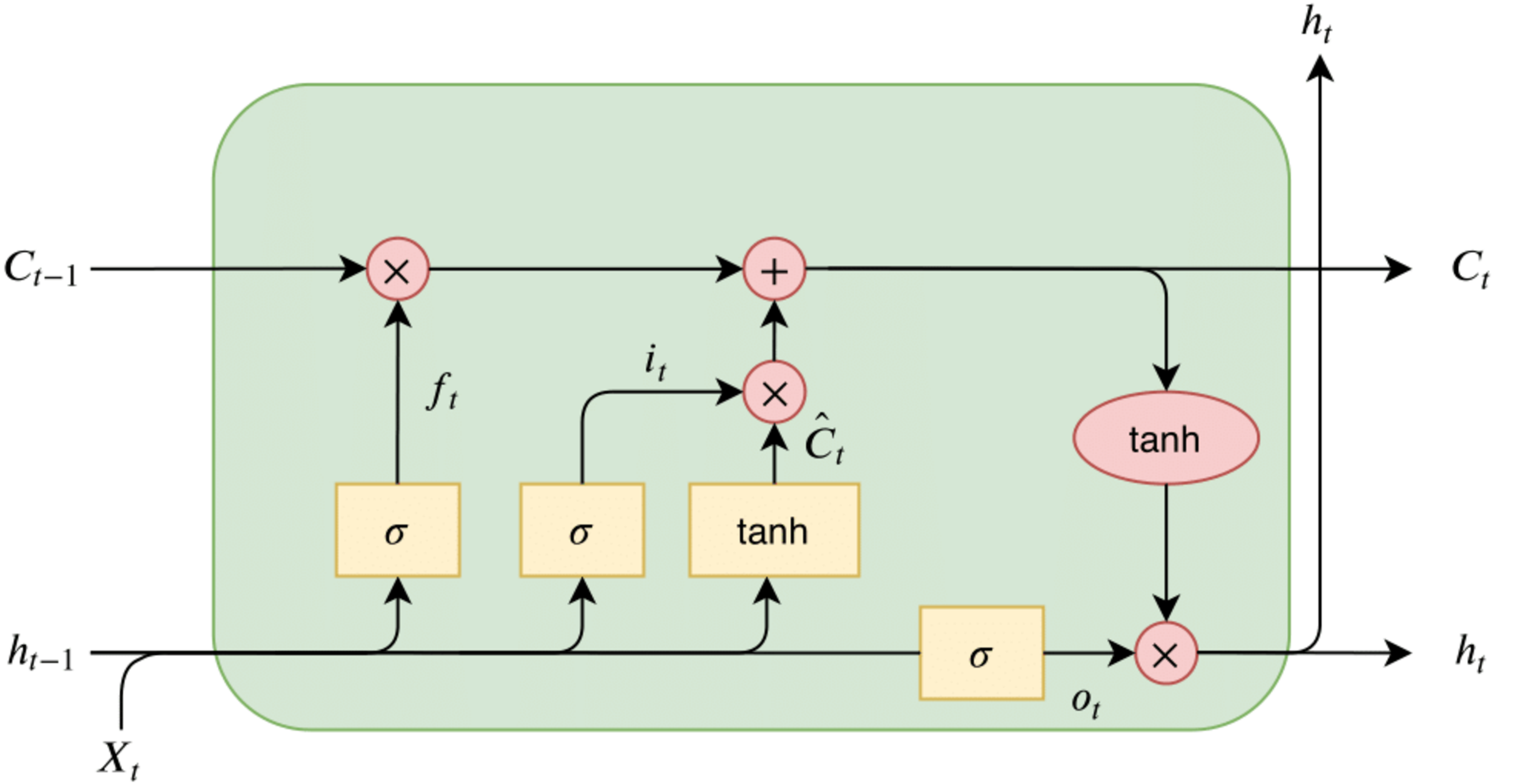
our feature selection method ?

Time Series approach :

TO TEST IF WE CAN MODEL THIS PROBLEM AS A TIME SERIES






h_t
C_{t-1}
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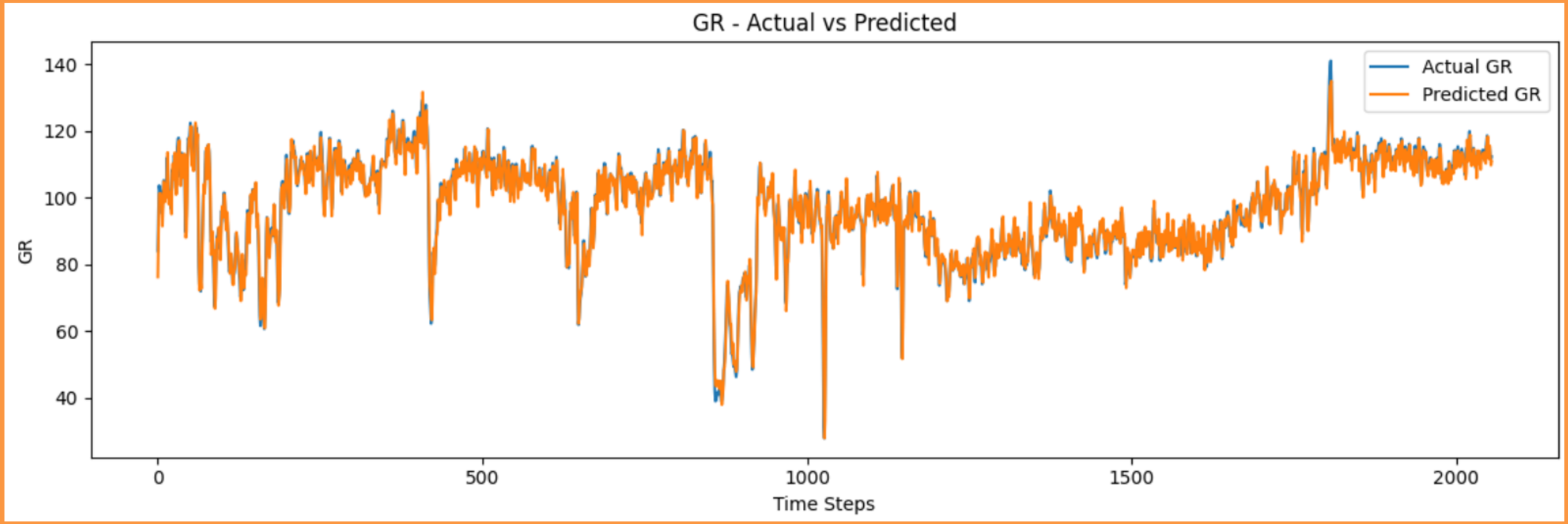


- **Mean Squared Error: 2.693241265688282**
- **Mean Absolute Error: 0.8901235733007917**
- **R^2 Score: 0.9732764980106002**

conclusion :

while this study appears promising, our meeting with Mr. Tadjine highlighted a significant limitation. He explained that it would be impractical to generalize this model in real-world applications due to the ever-changing structure of the Earth's surface. As a result, this idea will not be implemented in practice.

Clustering approach

II. LITHOLOGY DETECTION IN WELL LOGS: CLUSTERING CLAYS AND SANDSTONE ROCKS USING DATA MINING.

Approach:

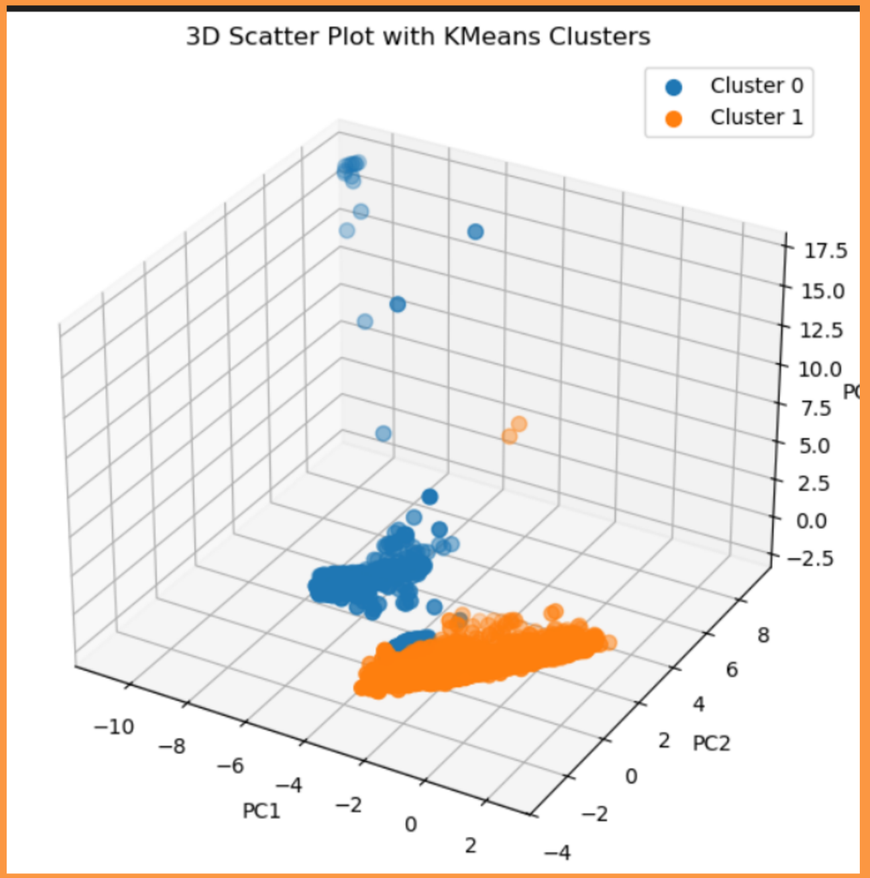
Using KMeans algorithm and DBSCAN to separate our data points into two clusters, the goal is not to only have a good separation but also to get insights of the lithology in the well.

Goal:

We should be able to have clear separation between Clays (Shales) and Sandstone, the latter that may indicate presence of oil.

KMeans Results:

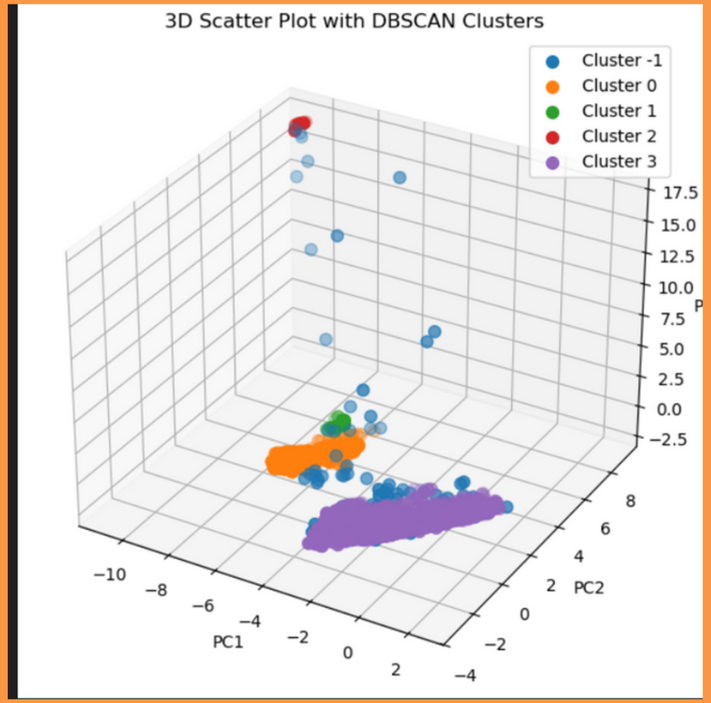
	GR					RHOB					...	PE					NPHI
	mean	std	min	50%	max	mean	std	min	50%	max	...	mean	std	min	50%	max	mean
Cluster_KMeans																	
0	22.932415	11.087212	9.8379	19.0323	106.7747	2.135175	0.186152	1.7157	2.1523	2.7380	...	8.661994	1.014561	4.4649	9.03490	9.5543	0.093248
1	92.078168	20.624413	11.8100	94.6883	170.5487	2.671668	0.062896	2.2197	2.6914	2.8551	...	4.072622	0.614973	2.2223	4.05015	8.5028	0.193946



- Matching results with theoretical value of the log data, can be improved.
- Some data points from cluster 0 should be included in in cluster 1, other data points are clearly outliers but they are considered in cluster 0; a common problem of KMeans algorithm.

DBSCAN Results:

	GR					RHOB					...	PE					NPHI	
	mean	std	min	50%	max	mean	std	min	50%	max	...	mean	std	min	50%	max	mean	s
Cluster_DB																		
0	20.738925	7.885715	9.8379	18.0746	45.1809	2.091675	0.153218	1.7157	2.1386	2.4220	...	9.066104	0.208724	8.4138	9.08650	9.5543	0.075794	0
3	91.679615	21.338932	11.8100	94.6050	170.5487	2.669949	0.067049	2.3017	2.6911	2.8551	...	4.080207	0.623132	2.2223	4.05255	8.2620	0.193649	0



- Matching results with theoretical value of the log data.
- A clear separation between clusters and the presence of various outliers, suggesting presence of other lithologies.

Conclusion

Our analysis of the well log data revealed two major clusters that correspond to clay and sandstone formations, supported by their matching log characteristics and external geological data. We also spotted smaller clusters that suggest other types of rock.
Among the clustering methods we tried, DBSCAN worked best because it can effectively group points that share similar properties—perfect for distinguishing between clays and sandstones. These findings are a good starting point for more detailed geological studies and exploration planning. However, we will likely need to define more clusters in future analyses to capture the full range of rock types.

Estimating Delta T values from the other attributes :

Introduction :

Since we found out that we cannot obtain satisfactory results for our main task (clustering) we decided to look for alternative topics that can have an added value and benefit the SONATRACH company. We decided DT to be the feature we'll be detecting since it is expensive to get its logs based on the following research

https://link.springer.com/article/10.1007/s10708-021-10502-6

Features selection :

```
DT: 7.7604
DEPT: 1.2399
RT90: 0.6239
NPHI: 0.5672
PE: 0.5341
GR: 0.2935
MIN1: 0.0000
```

We used domain knowledge to choose the most promising features among the 34 we had.

We used peasron correlation metric and mutual information to select among the ones we filtered using domain knowledge

we found out that most relationships are non linear and that MIN1 the density isn’t related to the sonic log

Data cleaning

- We decided to drop the noise since it was less than 10% of the data

- No duplicated not missing values where found
  - What’s particular about the data is the presence of outliers that are’nt due to noise and that we cannot elimintate that’s why I used two scaling methods (standard scaling/robust scaling)for the data to do a comparative analysis for the two scaling techniques when we have a lot of outliers among the data

Model selection

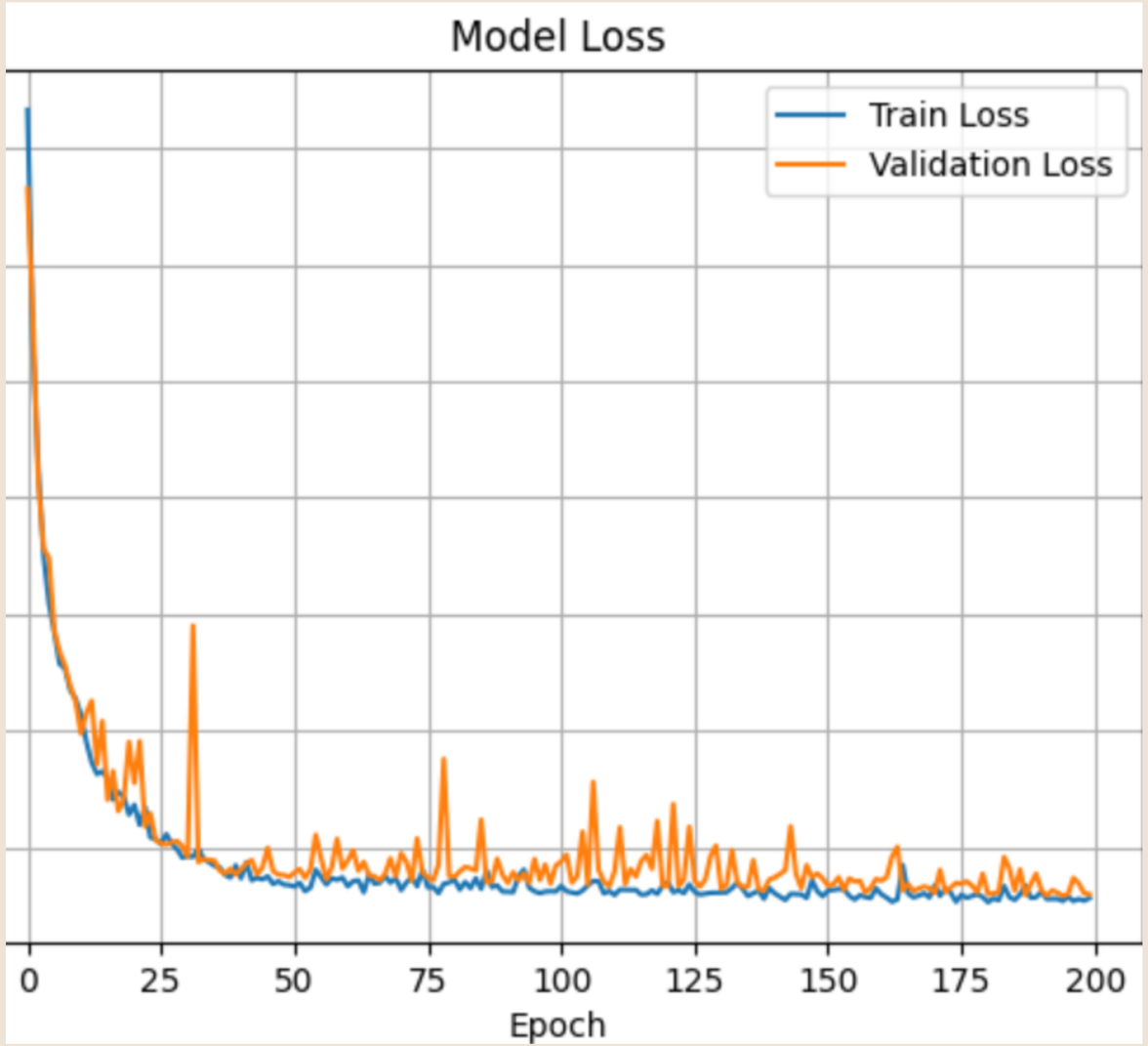
We decided to use ANN for the following two reasons:

- Features don't have a linear relationship with our feature (according to the pearson coeff) so we use a neural network since neural networks are powerful because they can learn complex, non-linear relationships between features

- The approach used in the research paper was based on ANN so we decided to use the same approach to get at least satisfactory results

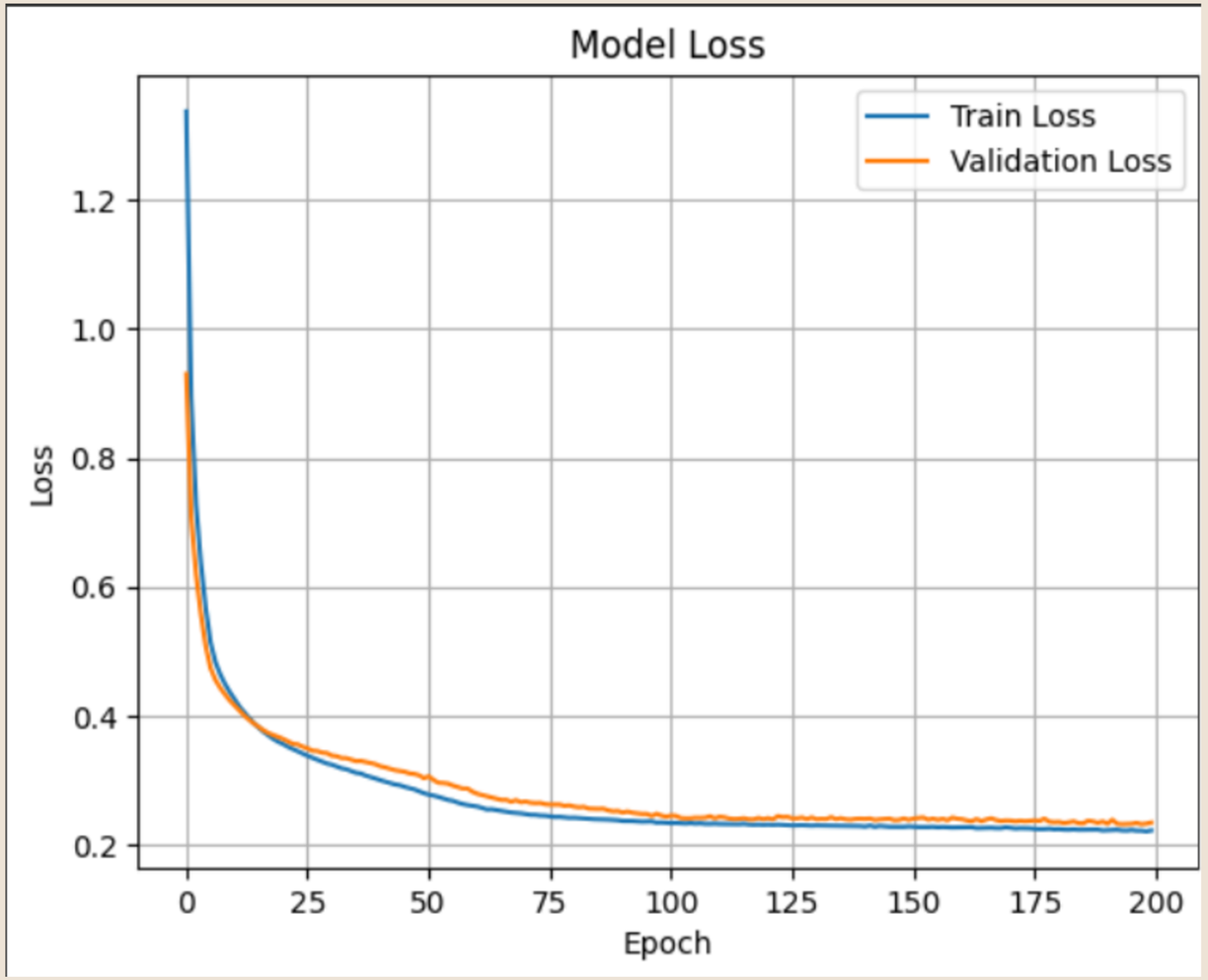
Results for the standard scaler

Inconsistency in the loss



Results for the robust scaler

The robust scaler uses the median and the quartile it gave more stable training curve



Conclusion

- it is better to use the robust scaler when there is an important presence of outliers in the dataset
- this topic is suitable for a time series study and not a regression study

Thank you !!

MEET THE TEAM MEMBERS :

- BOUZID NESRINE
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- KIHAL YASMINE
- BENMAICHE BESMALA RANDA

سوناطراك

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